# Paper 3 - D4/J3 Triality Surface

CQE-paper-03
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

## Construction Basis

Represent axis/sheet labeling, rotation/reflection equivalence, and Jordan carrier behavior as the first explicit triality surface.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

## Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-03.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-03\paper_kernel_manifest.json
- body: production\papers\CQE-paper-03\PAPER-BODY.md
- source: production\papers\CQE-paper-03\SOURCE.md
- formal: production\papers\CQE-paper-03\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-03\02-CQE-tool\TOOL.md
- tool_runner: production\papers\CQE-paper-03\02-CQE-tool\run.py
- workbook: production\papers\CQE-paper-03\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-03.md

# Paper 3 - D4/J3 Triality Surface

## Abstract

Paper 3 proves the first multi-language registration theorem in the CQECMPLX sequence. Paper 1 defines the local LCR carrier. Paper 2 identifies the first correction residue. Paper 3 proves that the same eight local states can be read without loss in three linked languages:

```
LCR state <-> D4-style axis/sheet code <-> diagonal J3 carrier
```

The theorem is intentionally local and finite. It does not claim the full D4 triality automorphism theorem, nor a full exceptional Jordan algebra action. It proves the registration surface needed by the next papers: a bijective axis/sheet encoding of `{0,1}^3`, trace preservation by `diag(L,C,R)`, and trace-2 diagonal idempotents for the shell-2 states.

## Claims

**Claim 3.1.** The eight binary LCR states admit a bijective D4-style axis/sheet chart.

**Claim 3.2.** The diagonal carrier `phi(L,C,R)=diag(L,C,R)` preserves the LCR shell as trace.

**Claim 3.3.** The shell-2 states map to trace-2 diagonal idempotents under coordinate-wise diagonal product.

**Claim 3.4.** The Paper 2 correction states are preserved as registered coordinates `(2,0)` and `(3,1)`.

## Definitions

Let:

```
S = {0,1}^3
s = (L,C,R) in S
shell(s) = L + C + R
```

Define the axis map:

```
axis(0,0,0) = 0    axis(1,1,1) = 0
axis(1,0,0) = 1    axis(0,1,1) = 1
axis(0,1,0) = 2    axis(1,0,1) = 2
axis(0,0,1) = 3    axis(1,1,0) = 3
```

Define the sheet map:

```
sheet(L,C,R) = 0 if L+C+R <= 1
sheet(L,C,R) = 1 if L+C+R >= 2
```

Define the diagonal carrier:

```
phi(L,C,R) = diag(L,C,R)
```

On diagonal carriers, use coordinate-wise diagonal product:

```
diag(a,b,c) o diag(a,b,c) = diag(a*a,b*b,c*c)
```

## Theorem 3.1 - Local Triality Surface

The map

```
tau(L,C,R) = (axis(L,C,R), sheet(L,C,R), diag(L,C,R))
```

is a faithful three-language presentation of the eight binary LCR states. The axis/sheet component is a bijection from `S` to `{0,1,2,3} x {0,1}`. The diagonal component preserves shell as trace. The shell-2 states map to trace-2 diagonal idempotents.

## Proof

The axis map partitions the eight states into four complement pairs:

```
axis 0: (0,0,0), (1,1,1)
axis 1: (1,0,0), (0,1,1)
axis 2: (0,1,0), (1,0,1)
axis 3: (0,0,1), (1,1,0)
```

Within each axis pair, one state has shell `0` or `1`, and the complement has shell `2` or `3`. The sheet bit therefore selects exactly one state inside each pair. Hence `(axis, sheet)` names exactly one state, and every state is named once.

For the diagonal carrier:

```
trace(phi(L,C,R)) = trace(diag(L,C,R)) = L + C + R = shell(L,C,R)
```

If `shell(s)=2`, then `phi(s)` has two diagonal `1` entries and one diagonal `0` entry. Since every binary entry is idempotent,

```
phi(s) o phi(s) = phi(s)
```

Thus shell-2 states become trace-2 diagonal idempotents. The Paper 2 correction states are among the enumerated states, and their registered coordinates are read directly from the axis/sheet chart:

```
(0,1,0) -> (2,0)
(1,1,0) -> (3,1)
```

This proves the local registration theorem.

## Receipt

The primary executable receipt is:

```
production/formal-papers/CQE-paper-03/verify_triality_surface.py
production/formal-papers/CQE-paper-03/triality_surface_receipt.json
```

The receipt status is `pass`. It verifies:

```
axis_sheet_bijection                    = true
axis_pairs_are_antipodal                = true
trace_equals_shell                      = true
shell2_states_are_diagonal_idempotents  = true
paper02_correction_coordinates_preserved = true
strong_triality_left_as_obligation      = true
```

## Scope Boundary

This paper proves a local triality surface. It does not, by itself, prove:

```
full D4 triality automorphism
full F4 action on J3(O)
off-diagonal octonionic dynamics
S3 group-ring closure of a Rule 30 trace-2 transition matrix
```

Those claims may be imported only when their own receipts are attached.

## Falsifiers

The paper fails if any of the following occur:

```
two different LCR states share the same axis/sheet coordinate
any LCR state lacks an axis/sheet coordinate
trace(diag(L,C,R)) differs from L+C+R
a shell-2 diagonal carrier is not idempotent
the Paper 2 correction states do not map to (2,0) and (3,1)
full D4/F4/J3(O) claims are counted as closed by this receipt alone
```

## Role in the Suite

Paper 3 supplies registered object transport. Paper 2 gives the correction residue, Paper 3 gives that residue a second coordinate language, and Paper 4 uses the registered residue as a boundary repair input.

The transfer is:

```
correction residue -> registered coordinate -> boundary repair constraint
```

## Open Obligations

- Wire `verify_triality_surface` into the installable kernel/API interface.
- Add the stronger group-action proof as a separate theorem if full D4/F4

language is used as proof rather than orientation.

- Reconcile all D-drive copies of the axis/sheet codec against this receipt.
- Ensure Paper 4 consumes the exact Paper 2 coordinates preserved here.

## Conclusion

Paper 3 proves that the local event can be registered without losing its state. The same `(L,C,R)` carrier can be read as an axis/sheet chart entry and as a diagonal trace carrier. That is the object-level registration discipline that later papers generalize into repair, path transport, and causal proof edges.